Sem IV	OE	Open Elective 1 (From Other School) Open-Source Technologies Introduction to Data Science	3	0	0	3	3
	PC	Design Thinking	0	0	2	2	1
	PC	Probability and Statistics Theory	3	0	0	3	3
	PC	Computer Organization and Architecture	3	1	0	4	4
	PC	Software Engineering	3	0	0	3	3
	PC	Software Engineering Laboratory	0	0	2	2	1
	PC	Theory of Computation	3	1	0	4	4
	PC	Design and Analysis of Algorithm	3	0	0	3	3
	PC	Design and Analysis of Algorithm Laboratory	0	0	2	2	1
			18	**2**	**6**	**24**	**23**
Sem V	OE	Open Elective 2 ((NPTEL/SWAYAM/MOOC))	3	0	0	3	3
	ESC	Introduction to Artificial Intelligence (Department Specific)	3	0	0	3	3
	HSC	Engineering Economics	3	0	0	3	3
	PE	Program Elective 1	3	0	0	3	3
	PC	Computer Networks	3	0	0	3	3
	PC	Computer Networks Laboratory	0	0	2	2	1
	PC	Compiler Design	3	0	0	3	3
	PC	Compiler Design Laboratory	0	0	2	2	1
	PC	Operating System	3	0	0	3	3
	PC	Operating System Laboratory	0	0	2	2	1
			21	**0**	**6**	**27**	**24**
Sem VI	OE	Open Elective 3 (From Other Department of FoET) Web Development Object Oriented Programming & Design	3	0	0	3	3
	PE	Program Elective 2	3	0	0	3	3
	PE	Program Elective 3	3	0	0	3	3
	PC	Cryptography and Network Security	3	0	0	3	3
	PC	Cryptography and Network Security Laboratory	0	0	2	2	1
	PC	Web & Mobile Development Essentials	1	0	0	1	1
	PC	Web & Mobile Development Essentials Laboratory	0	0	4	4	2
	PC	Distributed Computing	3	0	0	3	3
	PC	Distributed Computing Laboratory	0	0	2	2	1
			16	**0**	**8**	**24**	**20**
Sem VII	Pro	Summer Internship	0	0	0	0	2
	OE	Open Elective 4 (From Other Department of FoET) Green IT	3	0	0	3	3

		Global Data Security & Privacy Laws					
	ESC	Industry 4.0 (Department Specific)	2	0	0	2	2
	ESC	Industry 4.0 Laboratory (Department Specific)	0	0	2	2	1
	PE	Program Elective 4	3	0	0	3	3
	PE	Program Elective 5	3	0	0	3	3
	PC	Machine Learning	3	0	0	3	3
	PC	Machine Learning Laboratory	0	0	2	2	1
	PC	Cyber Laws and Ethics	1	0	0	1	1
	Pro	Seminar					1
			15	**0**	**4**	**19**	**20**
Sem VIII		**Course Name**	**Th**	**Tut**	**Pra**	**Hrs**	**Cr**
	Pro	Major/Comprehensive Project					**12**

Total Credits: 160

List of Electives (Track-based)

	Artificial Intelligence/ Machine Learning Track	**Industry Track**	**Speech, Vision & Text Track**	**Computing Track**	**Communication Track**
PE-1	Data Mining and Data Warehousing	Object Oriented Modelling and Design	Computer Graphics	Advanced Data Structure and Algorithms	Data Communication
PE-2	Soft Computing	UI/UX Design	Digital Image Processing	Blockchain Technology	Mobile Computing
PE-3	Deep Learning	Secure Software Engineering	Computer Vision	Big Data Analytics	Wireless Sensor Networks
PE-4	Agent based Learning	Agile and DevOps	Natural Language Processing	Cloud Computing	Internet of Things
PE-5	Machine Learning in Cyber Security	Web Application Testing	Speech Processing	Quantum Computing	Autonomous Systems

5th Semester

24CS301T					Introduction to Artificial Intelligence					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25			100

COURSE OBJECTIVES

- To identify the problems where AI is required and the different methods available.
- To compare different AI techniques available.
- To define and explain learning algorithms and identify problems in game playing.
- To learn Neural Networks and Expert systems

UNIT 1: Introduction To Ai And Searching AI Problems, Intelligent Agents, Types of Learning, Problem Formulation, Basic Problem-Solving Methods. Search strategies, Uniformed Search Strategies, State-Space Search, BFS, DFS, Greedy Best First Search, A* Search, Hill Climbing, and Adversarial Search: minimax and alpha-beta pruning.	**12 Hrs.**
UNIT 2: Foundations And Applications Of Machine Learning Techniques Machine Learning: Representation (features), Linear Transformations, Decision Tree Classifiers, Introduction to Random Forest, Linear Regression, Problem formulations (classification and regression), PCA and Dimensionality Reduction, Nearest Neighbours and KNN, Applications of Machine Learning.	**10 Hrs.**
UNIT 3: Neural Networks: Structure, Learning, And Applications Characteristics of Neural Networks, Artificial Neural Networks: Terminology, Models of Neuron, Topology, Basic Learning Laws, Back Propagation and Gradient Decent, Regularization, Applications of Deep Learning.	**12 Hrs.**
UNIT 4: Introduction To Reinforcement Learning Markov Decision Processes, Introduction to Reinforcement Learning, Policy Search, Dynamic Programming & TD Methods, Bellman Optimality, Q-Learning, SARSA, Applications of Deep Reinforcement Learning.	**8 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Demonstrate understanding of foundational concepts in artificial intelligence and problem-solving methods.

CO2 - Analyse different informed, uninformed and adversarial search techniques in Artificial Intelligence.

CO3 - Understand basics of machine learning including supervised and unsupervised techniques, such as decision tree classifiers, Nearest Neighbours and Linear Regression.

CO4 - Understand of the basics of Neural Networks, including their architecture, components, and applications in AI.

CO5 - Comprehend basics of Reinforcement Learning, policy search, Dynamic programming and Reinforcement Learning algorithms.

CO6 - Analyse real world applications in Machine Learning, Deep Learning and Deep Reinforcement Learning.

TEXT/REFERENCE BOOKS

1. Russell, S.J. and Norvig, P., Artificial Intelligence: A Modern Approach, Pearson Education.
2. Tom Mitchell, Machine Learning: An Artificial Intelligence Approach
3. Reinforcement Learning: An Introduction Book by Andrew Barto and Richard S. Sutton
4. Kevin Night and Elaine Rich, Nair B., “Artificial Intelligence (SIE)”, McGraw Hill.
5. Dan W. Patterson, “Introduction to AI and ES”, Pearson Education.
6. G.Luger, W.A. Sttubblefield, “Artificial Intelligence”, Addison-Wesley Longman.

24CS302T					Computer Networks					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To study the TCP/IP layered architecture and various layer functionalities.
- To understand the different routing strategies in the computer networks.
- To understand the error control and flow control mechanism in TCP/IP network.
- To understand the application layer protocols for various networking applications.

UNIT 1: Network Stack and Data Link Layer Introduction to Computer Network and Internet, Layered Architecture (OSI and TCP/IP), Guided and Unguided Media, Framing, Flow Control , Error Control, Media Access Control (ALOHA, CSMA/CD), Ethernet 802.3, Wireless LAN 802.11, Token Ring 802.5, Reliability Issue, Sliding Window.	**13 Hrs.**
UNIT 2: Network Layer Internetworking and Routing: Best effort Service, Switching, Virtual Circuit, IP Addressing, Routing Issue, Distance Vector and Link State Routing, Open Shortest Path First, Routing Information Protocol, Border Gateway Protocol, IPv6 Addressing, Protocol, Transition from IPv4 to IPv6.	**12 Hrs.**
UNIT 3: Transport Layer End to end delivery issues, Reliable data transfer, Congestion control, Traffic engineering and Quality of Service, TCP and UDP.	**11 Hrs.**
UNIT 4: Application Layer Introduction to Application Layer, Application layer protocols: Domain Name System, File Transfer Protocol, Hyper Text Transfer Protocol, Simple Mail Transfer Protocol, Multipurpose Internet Mail Extension.	**06 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the functionalities of OSI and TCP/IP model.
CO2 - Discuss the MAC layer protocol and LAN technologies.
CO3 - Analyze the Distance Vector and Link State Routing protocol strategies in IP network.
CO4 - Compare the IPv4 and IPv6 Addressing schemes.
CO5 - Determine the congestion control mechanism in TCP/IP network.
CO6 - Examine the working and architecture of client server application.

TEXT/REFERENCE BOOKS

1. Andrew S Tenenbaum, “Computer Networks” Pearson Education.
2. Behrouz A Forouzan, “Data Communication and Networking”, McGraw Hill.
3. William Stalling, “Data and Computer Communication”, Pearson Education
4. James Kurose and Keith Rose, “Computer Networking: A Top Down Approach” Pearson Education
5. Pearson Larry Peterson and Bruce Davie, Computer Networks, A Systems Approach, Morgan Kaufmann

24CS302P					Computer Networks Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

PREREQUISITE: Computer Programming Language, Data Structure

COURSE OBJECTIVES

- To design the local area networking topologies
- To analyze the traffic load for network architecture
- To understand the working of interior and exterior routing protocols
- To design client server application using socket programming

LAB EXPERIMENTS

- **Network Devices**
  Hubs, L-2 and L-3 Switches, Router, Repeater, Transmission Media
- **Network Topologies Design**
  Network Topologies – Mesh, Star, Bus, Ring, Hybrid
- **Sliding Window Protocol**
  Stop and Wait, Go Back N, Selective Repeat
- **Framing**
  Bit Stuffing, Byte Stuffing, Escape Sequence
- **Error Control**
  Single Bit Error, Burst Error, Checksum, Cyclic Redundancy Check
- **Virtual Local Area Network**
  Virtual Local Area Network, Traffic Load Estimation
- **Loop Avoidance Techniques**
  Spanning Tree Protocol, Rapid Spanning Tree Protocol
- **Interior Routing Protocol**
  Open Shortest Path First, Routing Information Protocol
- **Exterior Routing Protocol**
  Border Gateway Protocol
- **Socket Programming**
  Transmission Control Protocol Socket, User Datagram Protocol Socket.
- **Application Layer Protocol**
  Domain Name System, Hyper Text Transfer Protocol, Simple Mail Transfer Protocol, File Transfer Protocol

COURSE OUTCOMES

On completion of the course, student will be able to
CO1 - Understand the functionalities of networking devices.
CO2 - Demonstrate the working of LAN topologies and virtual local area network.
CO3 - Construct the loop avoidance mechanism in local area network.
CO4 - Implement the interior and exterior routing protocol strategies.
CO5 - Analyze the traffic load for networking protocols.
CO6 - Design client server application using TCP and UDP port.

TEXT/REFERENCE BOOKS

1. Kenneth L. Calvert, Michael J. Donahoo, “TCP/IP Sockets in Java: Practical Guide for Programmers”, Morgan Kaufmann
2. Elliotte Rusty Harold, “Java Network Programming: Developing Networked Applications”, O'Reilly Media
3. James H. Baxter, “Wireshark Essentials”, PACKT Publishing

24CS303T					Compiler Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

PREREQUISITE: Concepts of data structures, Theory of computation, Programming skills

COURSE OBJECTIVES

- Familiarize with the front-end as well as back-end stages of compiler design, source file, object file and executable file structures and libraries.
- Understand the differences between constructing lexers/parsers by hand
- Get an experience with generating intermediate representations and learn the importance of designing simpler languages
- Observe how original source program is transformed to an optimized target code using code optimization techniques

UNIT 1: Introduction and Lexical Analysis Overview of compiler, Introduction of phases of a compiler, Cousins of the Compiler- Brief discussion on Preprocessor, Assembler, Linker, Loader, Front-end and Back-end of compiler, Lexical analysis and representation, Input Buffering, Token Recognition and Finite Automata, Error Recovery of lexical analysis.	**10 Hrs.**
UNIT 2: Syntax Analysis Introduction, Parsing, Role of Parsers, Context Free Grammars, Top-Down Parsers (RDP, LL), Bottom-Up Parsers (Operator-Precedence, LR), Error recovery strategies	**13 Hrs.**
UNIT 3: Syntax Directed Translation and Intermediate Code Generation Syntax Directed Definition, S-attributed and L-attributed Definitions, Evaluation order, Syntax directed definition and translation schemes, Different Intermediate Forms, Implementation of Three Address Code, Intermediate code for basic constructs of languages	**10 Hrs.**
UNIT 4: Runtime Environments, Code Generation and Optimization Storage allocation strategies, Parameter passing, Type checking, Data flow analysis- Basic blocks and flow graphs, liveliness analysis, local and global optimization techniques- semantic preserving transformations and loop optimizations, Target code generation	**09 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Discuss all phases of compiler and roles of lexical analysis.
CO2 - Understand top-down and bottom-up parsing algorithms and error recovery strategies.
CO3 - Illustrate syntax directed translations and evaluation order in parse trees.
CO4 - Analyse the forms of intermediate representations of a source program during compilation.
CO5 - Identify storage allocation and data flow between procedures.
CO6 - Recognize program transformation techniques and generation of target code.

TEXT/REFERENCE BOOKS

1. Alfred V Aho, M S. Lam, R Sethi, Jeffrey D. Ullman. Compilers-Principles, Techniques and Tools, Pearson.
2. Steven Muchnick, Advanced Compiler Design and Implementation, Morgan Kaufmann.
3. K. Muneeswaran,Compiler Design, Oxford university press.
4. Keith D. Cooper and Linda Torczon. "Engineering a Compiler", Morgan Kaufmann.

24CS303P					Compiler Design Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
-	-	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- Familiarize with the front-end as well as back-end stages of language processors
- Understand the differences between constructing lexers/parsers by hand versus using automated generators
- Relate the applications of data structures and theory of computation in compiler design
- Hands-on experience with generating intermediate representations for simpler languages
- Meet the requirements of the realistic constraints of compilers while generating target code

LIST OF EXPERIMENT

1. **Lexical analysis using higher level programming:**
   Program to identify keywords, identifiers and others from the given input file using higher level programming.

2. **Lexical Analysis Tool:**
   Programs based on lexical analysis tool (flex or similar).

3. **Top-down parser:**
   Write a program to implement Recursive Decent Parser (RDP) or similar for given grammar.
4. **Bottom-up parser:**
   Program to implement any one shift-reduce parser.
5. **YACC tool:**
   Programs based on YACC tool and context free grammars.

6. **Intermediate code generation:**
   Program related to intermediate code generation from given expression.
7. **Target code generation:**
   Program to generate target code from the intermediate code.

8. **Demo compiler:**
   Implement a demo compiler.

COURSE OUTCOMES

On completion of the course, student will be able to
CO1 - Identify lexemes and tokens in the given input string using higher level programming language.
CO2 - Demonstrate lexical analysis tool for validation of strings accepted by regular expressions.
CO3 - Experiment top-down and bottom-up parsers for comparative study.
CO4 - Explore parsing tool to derive the strings from context free grammars.
CO5 - Build appropriate intermediate code for programming constructs.
CO6 - Apply code optimization techniques related to target code generation.

TEXT/REFERENCE BOOKS

1. John R. Levine, Tony Mason, Doug Brown, lex & yacc, O'Reilly.
2. AW Appel, J Palsberg, Modern Compiler Implementation in Java, Cambridge University Press.
3. AW Appel, M Ginsburg, Modern Compiler Implementation in C, Cambridge University Press.

24CS304T					Operating System					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

PREREQUISITE: Basic C Programming and Data Structures

COURSE OBJECTIVES

- To understand the Operating System (OS) role in the overall computer system and to study the operations performed by OS as a resource manager.
- To understand the scheduling and memory management protocols by OS for processes.
- To understand physical and logical memory system within OS and its management algorithms.
- To understand the concepts of Deadlock, input/output, storage and file management by the operating system.

UNIT 1 Introduction of Operating System and Process Management Operating System Objectives and Functions, Computer-System Organization, Computer-System Architecture, Operating-System Operations, Operating-System Structure, Process Management, Memory Management, Storage Management, Protection and Security, Operating-System Structure, System calls and its Types, Process and Threads, Types of Threads, multicore and multithreading, Process Concept, Operations on Process, CPU Scheduling-Basic concepts and algorithms, Thread scheduling.	**10 Hrs.**
UNIT 2 Synchronization Principles of Concurrency, Interprocess Communication: shared and message passing., Race Condition, Mutual Exclusion, the critical section problem, Peterson's Solution, Fork() and Join(), Synchronization Hardware Support, Mutex locks, Semaphores: Usage and implementation, Classical Problems of Synchronization: Bounded-Buffer Problem, Readers–Writers Problem, Dining-Philosophers Problem, Monitors and its usage in Classical Synchronization Problems.	**10 Hrs.**
UNIT 3 Deadlocks Deadlock Characterization, Deadlock Prevention: Ostrich Algorithm, Deadlock Avoidance: Banker's Algorithm, Deadlock Detection, and Recovery from Deadlock: Resource allocation graph and wait-for-graph.	**10 Hrs.**
UNIT 4 Memory and File Management Memory Management: Address Binding, Swapping, Contiguous Memory Allocation, Segmentation, Paging. Virtual Memory Management: Background, Demand Paging, Copy-on-Write, Page Replacement, Page Replacement Algorithms, Allocation of Frames, Thrashing. Storage Management- File and Directory structure, Unix I-node Structure, Disk Structure, Disk Attachment, Disk Scheduling.	**12 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 - Recognize the fundamental concepts related to CPU management aimed at enhancing system performance.
CO2 - Understand synchronization techniques to manage concurrent processes effectively within the operating system environment.
CO3 - Understand the memory organization in the CPU and its management by the operating system.
CO4 - Assess turnaround time, waiting time, and response time to conduct a comparative analysis between CPU scheduling algorithms.
CO5 - Apply Banker's algorithm as a preventive measure against deadlock situations within the operating system.
CO6 - Assess disk scheduling strategies to efficiently manage the file system operations within the operating system environment.

TEXT/REFERENCE BOOKS

1. Abraham Silberschatz, Peter B. Galvin, Greg Gagne, Operating System Concepts, Wiley.
2. W. Stallings, Operating Systems – Internals and Design Principles, Prentice Hall
3. Sumitabha Das, Unix Concepts and Applications, TMH.
4. Andrew S Tanenbaum, Modern Operating Systems, PHI

24CS304P					Operating System Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

PREREQUISITE: Basic C Programming and Data Structures

COURSE OBJECTIVES

- To gain hands-on experience in implementing and understanding system calls in a Unix-like environment.
- To become proficient in navigating the Ubuntu Linux operating system environment.
- To familiarize with the fundamentals of shell scripting using Bash on Ubuntu, including script creation, execution permissions, variables, control structures, and input/output redirection.

LIST OF EXPERIMENTS:

1. **Implementation of Linux basic commands on Unix/Linux**
   - cal, echo, date, ls, lp, man, who, whomi, unname, hostname, bc, uname, uptime, help, cd, mkdir, ls -ltr, bc -l, Unix editor commands: vi, touch, nano, File manipulation commands: cat, grep, rm, touch, cp, mv, cut, head, tail, chmod, wc
2. **Implementation of Shell fundamentals on Unix/Linux**
   - If Else construct, looping concepts, switch case concepts, user defined function and the recursion concepts
3. **Programming simple shell programs using shell fundamentals.**
   - To check the given number is even or odd.
   - To check the given year is leap year or not.
   - To find the factorial of a number.
   - To swap the two integers.
   - To create user defined functions for computing minimum and maximum values from the input by the user using command line arguments.
4. **Implementation of system calls of UNIX operating system in C**
   - fork, exec, getpid, exit, wait, close, stat, opendir, readdir
5. **Simulation of unix commands**
   - cp, ls, grep, rm
6. **Group projects**

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall and list basic Linux commands and their functionalities.

CO2 - Explain the working principles of shell fundamentals.

CO3 - Develop and implement simple shell programs to accomplish specific tasks such as checking for even/odd numbers, leap years, factorial calculation, and integer swapping.

CO4 - Implement system calls of UNIX operating system in C programs to create processes, manage files, and retrieve directory information.

CO5 - Implement existing system calls of UNIX system in C language.

CO6 - Design and collaboratively work on group projects to integrate process synchronization, scheduling and deadlock concepts learned throughout the theory sessions and produce functional software artifacts.

TEXT/REFERENCE BOOKS

1. Sumitabha Das, Unix Concepts and Applications, TMH.
2. W. Stallings, Operating Systems – Internals and Design Principles, Prentice Hall
3. Ubuntu Unlished, Matthew Helmke, Addison-Wesley Professional

Program Elective Course Track 1

Artificial Intelligence/ Machine Learning Track

24CS331T					Data Warehousing and Data Mining					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

PREREQUISITES: Basic knowledge of linear algebra, probability, and statistics.

COURSE OBJECTIVES

- To understand the principles of Data warehousing and Data Mining.
- To be familiar with the Data warehouse architecture and its Implementation.
- To understand the concept of Analytical Processing (OLAP).
- To characterize the kinds of patterns that can be discovered by association rule mining, classification and clustering.

UNIT 1: Introduction To Data Warehousing And Data Mining Data Warehouse Definition, Multidimensional Data Models, Data Warehouse Architecture, Data Granularity, Information Flow Mechanism, Two Classes of Data, Data Lifecycle, Data Flow From Warehouse to Operational Systems, Data Mining Functionalities, Classification of Data Mining System, Data Warehouse v/s Data Mining, Data Mining Process, Major Issues in Data Mining, Knowledge Discovery in Databases (KDD), KDD Process, Online Analytical Processing (OLAP)– OLAP and Multidimensional Data Analysis, On-Line Analytical Processing And Mining	**13 Hrs.**
UNIT 2: Data Mining – Preprocessing Descriptive Data Summarization, Data Cleaning, Integration and Transformation, Data Compression and Dimension Reduction, Principal Component Analysis, Binning Methods, Data Reduction Techniques: Singular Value Decomposition (SVD), Discrete Fourier Transform (DFT), Discrete Wavelet Transform (DWT), Data Discretization and Concept Hierarchy Generalization	**08 Hrs.**
UNIT 3: Association Analysis Basic Concepts, Frequent Item Set Generation, Rule Generation, Various Kinds of Association Rules, Market Basket Analysis, Apriori Algorithm, Eclat Algorithm, FPMax Algorithm, FP-Growth Algorithm, Classification by Decision Tree, Bayesian Classification, Rule Based Classification, Classification by Back propagation, Support Vector Machines, Lazy Learners, Prediction Techniques, Effect of Skewed Support Distribution, Multilevel and Multidimensional, Association Rule Mining v/s Correlation Analysis, Constraint-Based Association Mining.	**12 Hrs.**
UNIT 4: Data Mining On Complex Data And Applications Clustering Algorithms, Cluster Analysis, Algorithms for mining of spatial data, social impacts of data mining, recent trends in data mining, Web Content Mining, Web Structure Mining, Web Usage Mining, Spatial Mining, Multimedia Data Mining, Text Mining,	**09 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of this course, students will be able to:

CO1 - Understand the fundamental concepts of data mining and data warehousing.

CO2 - Identify the need of Data Warehouse tools and techniques for designing and developing different types of databases.

CO3 - Preprocess data for mining by handling missing values, noise, and outliers.

CO4 - Describe the use of Online Analytical Processing to analyze and interpret data

CO5 - Apply various data mining techniques, including association rule mining, classification, clustering, and anomaly detection.

CO6 - Evaluate and interpret the results of data mining algorithms.

Program Elective Course Track 2

Industry Track

24CS332T					Object Oriented Modelling and Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- Understand Object-Oriented Modelling and Design (OOMD) principles and contrast them with traditional modelling approaches.
- Appraise the appropriate use of OOP concepts in program design within the context of OOMD.
- Get acquainted with advanced OOP concepts such as streams and multi-file programs, exceptions, and standard template library, and understand their relevance in OOMD.
- Understand the use of performance optimization techniques in OOMD to enhance system efficiency.

UNIT 1: Introduction, Modelling Concepts and Class Modelling: Overview of Object-Oriented Modelling and Design (OOMD) principles, Contrasting OOMD with traditional modelling approaches, Understanding the core concepts of OOMD, Modelling Fundamentals: Object and Class Concept in modelling, Introduction to associations and relationships, Encapsulation and abstraction in modelling, Access control and visibility modifiers, Constructors and Destructors in modelling, Introduction to Inheritance and its significance in design, Navigation of class models; Advanced Class Modelling, Advanced object and class concepts; Association ends; N-ary associations; Aggregation; Abstract classes; Multiple inheritance; Metadata; Reification; Constraints; Derived Data; Packages.	**12 Hrs.**
UNIT 2: Use Case Modelling and Detailed Requirements: Comprehensive Overview; Elaborate Object-oriented Requirements Definitions; System Processes: A Detailed Use Case/Scenario Perspective; Identifying Input and Outputs: The System Sequence Diagram; Identifying Object Behaviour: The State Chart Diagram; Integrated Object-oriented Modelling Approach.	**10 Hrs.**
UNIT 3: Process Overview, System Conception, and Domain Analysis: Comprehensive Overview; Development Stages and Life Cycle; System Conception: Formulating a System Concept; Elaborating the Concept; Crafting a Problem Statement. Domain Analysis: Comprehensive Analysis Overview; Domain Class Model: Detailed Domain State Model; Domain Interaction Model; Iterative Analysis Process.	**10 Hrs.**
UNIT 4: Use Case Realization: Design Discipline in Iterative Development; Object-Oriented Design: Bridging Requirements and Implementation; Design Classes and Class Diagrams; Interaction Diagrams: Implementing Use Cases and Defining Methods; Designing with Communication Diagrams; Enhancing the Design Class Diagram; Package Diagrams: Organizing Major Components; Implementation Considerations for Three-Layer Design.	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand object-oriented programming principles.

CO2 - Evaluate different object-oriented modeling techniques and assess their suitability for specific software development scenarios.

CO3 - Develop comprehensive object-oriented models for real-world systems using appropriate design patterns.

CO4 - Critically analyze existing object-oriented systems to identify design flaws and propose improvements.

CO5 - Assess the effectiveness of various object-oriented design patterns in promoting modularity, reusability, and maintainability.

CO6 - Integrate object-oriented design principles with other software engineering paradigms to create holistic software architectures that meet both functional and non-functional requirements.

TEXT/REFERENCE BOOKS

1. "Object-Oriented Analysis and Design with Applications" by Grady Booch, Robert A. Maksimchuk, Michael W. Engle, Bobbi J. Young, Jim Conallen, and Kelli A. Houston
2. "Design Patterns: Elements of Reusable Object-Oriented Software" by Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides
3. "Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development" by Craig Larman
4. "Object-Oriented Modeling and Design with UML" by Michael Blaha and James Rumbaugh

Program Elective Course Track 3

Speech, Vision & Text Track

24CS333T					Computer Graphics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To introduce concept of computer assisted picture generation and manipulation
- To understand an overview of interactive computer graphics, two-dimensional system.
- To understand the most important algorithm for graphical primitives, transformation, clipping and filing for 2D objects
- To study curve generation, 3D picture generation, transformation and animation techniques

UNIT 1: Basic of Computer Graphics and Primitive Drawing Algorithms Basics of Computer Graphics, Applications of computer graphics, Display devices, Graphics input devices, basic primitive drawing algorithms- point, line, circle, ellipse, and polygon drawing algorithms, polygon filling algorithms	**11 Hrs.**
UNIT 2: Two-dimensional Transformation and Viewing Two-dimensional transformations (translation, rotation, scaling, reflection, shearing), matrix representation, composite transformations, homogeneous coordinates, viewing transformation, line clipping, polygon clipping	**11 Hrs.**
UNIT 3: Three-dimensional Object Representations and Transformations Three-dimensional object representations, Three-dimensional transformations (translation, rotation, scaling, reflection, shearing), projection and types of projection, projection transformations, modelling curves, curved lines and surfaces, Visible surfaces	**11 Hrs.**
UNIT 4: Rendering and Colour Models Polygon Rendering methods, Texture mapping, Colour models (RGB, YIQ, CMY), Modelling techniques and fractals, Animation: Computer assisted animation and real time animation techniques	**09 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Discuss the concepts of computer graphics and hardware devices.
CO2 - Compare the algorithms for 2D graphics primitives and attributes.
CO3 - Apply composite geometric transformations and viewing on 2D objects.
CO4 – Represent 3D objects with transformations and projections.
CO5 - Demonstrate the concepts of clipping and visible surface detection in 2D and 3D viewing.
CO6 - Differentiate rendering techniques, colour models and animation techniques for object representation.

TEXT/REFERENCE BOOKS

1. John F. Hughes, Andries van Dam, Morgan McGuire, David F. Sklar, James D. Foley, Steven K. Feiner, Kurt Akeley. Computer Graphics: Principles and Practice, Pearson education
2. David F. Rogers, Mathematical elements for computer graphics, Tata McGraw Hill
3. Donald Hearn, Pauline Baker, Computer graphics with OpenGL, Pearson education

Program Elective Course Track 4

Computing Track

24CS335T					Advanced Data Structures and Algorithms					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Understand and implement advanced data structures such as AVL trees, B-trees, red-black trees, Fibonacci heaps, and suffix trees, etc.
- Choose appropriate data structures to design algorithms for a specific problem.
- Develop proficiency in designing and analysing algorithms for complex problems
- Gain hands-on experience through projects or case studies that involve implementing and analysing algorithms and data structures in practical scenarios.

UNIT 1: Hashing And Skip Lists Review of Hashing, Hash Function, Collision Resolution Techniques in Hashing, Separate Chaining, Open Addressing, Linear Probing, Quadratic Probing, Double Hashing, Rehashing, Extendible Hashing Search and Update Operations on Skip Lists, Probabilistic Analysis of Skip Lists, Deterministic Skip Lists	**10 Hrs.**
UNIT 2: Advanced Trees Revision to BST and AVL trees, Red Black Trees, 2-3 Trees, B-Trees, Splay Trees, Priority Search Tree, Priority Range Trees, Quad trees, k-D Trees, Fibonacci heaps and suffix trees.	**10 Hrs.**
UNIT 3: Advanced Algorithms Maximum matching in graphs, Edmond's Blossom algorithm for augmenting path, Maxflow-mincut theorem, Ford-Fulkerson and Edmond-Karp maximum-flow algorithm, Chinese Remainder Theorem, DFT and FFT algorithms, Simplex method.	**12 Hrs.**
UNIT 4: Approximation And Randomized Algorithms Interior point method, ellipsoid method, sequential quadratic programming, Randomized Quicksort, Ant colony optimization, Monte-Carlo Method.	**10 Hrs.**
	42 Hrs

COURSE OUTCOMES

After completion of the course, students will be able to:
CO1 - Describe various collision resolution techniques in hashing.
CO2 - Understand properties and operations associated with advanced tree data structures.
CO3 - Apply advanced trees to design efficient data structures for specific applications, such as priority search trees or suffix trees.
CO4 - Analyse the time and space complexities of standard advanced algorithms.
CO5 - Evaluate the suitability of different approximation and randomized algorithms for solving specific optimization problems.
CO6 - Develop novel algorithms or variations of existing algorithms to address optimization problems, considering both deterministic and randomized approaches.

TEXT/REFERENCE BOOKS

1. Mark Allen Weiss - Data Structures and Algorithm Analysis in C++, Pearson.
2. M T Goodrich, Roberto Tamassia - Algorithm Design, John Wiley.
3. Charles E. Leiserson, Thomas H. Cormen, Ronald L. Rivest, Clifford Stein - Introduction to Algorithms, MIT Press.
4. Klienberg and Tardos, Algorithm Design, Pearson.

Program Elective Course Track 5

Communication Track

24CS334T					Data Communication					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand the importance of data Communication.
- To understand the fundamental communication concepts of digital and analog transmissions.
- To demonstrate digital signal transmission and encoding techniques.
- To illustrate the working of signal and carrier wave multiplexing techniques.

UNIT 1: Data Communication Overview and Physical Layer Introduction – Process of Data Communication, Transmitter, Receiver, Medium and Message Protocol, Protocols, Standards and Standard organizations, Bandwidth, Data Transmission Rate, Baud rate and Bits per second, modes of communication, OSI/TCIP Model, Networks, Internet, Intranet, Protocols, Telephone and Cable network, Physical Layer – Signals, Analog, Digital, Analog vs. Digital, Transmission impairment, Data Rate Limits, Noiseless and Noisy channel, Performance	**11 Hrs.**
UNIT 2: Data Transmission and Signal Encoding Techniques Analog Transmission – Analog to Analog conversion: Amplitude Modulation, Frequency Modulation, and Phase Modulation, Digital to Analog conversion: ASK, FSK, PSK, and QAM. Digital Transmission – Analog to digital conversion, Digitization, PCM, Transmission Modes, Line Coding (Unipolar, Polar, Biphase), Block Coding (4B/5B Encoding), DTE-DCE Interface, Modems, 56K Modems, Cable Modems.	**12 Hrs.**
UNIT 3: Transmission Media Transmission Media – Guided Media: Twisted Pair cable, Coaxial cable, and Fiber optic cable, Unguided Media: Radio Waves, Microwaves, and Infrared satellite, Line-of-site transmission: point-to-point and broadcast.	**08 Hrs.**
UNIT 4: Multiplexing Multiplexing – FDM, WDM, Synchronous TDM (time slots & frames, interleaving, data rate management), Statistical Time Division Multiplexing, Spread Spectrum – FHSS, DSSS, Switching: Circuit-Switched Networks, Datagram networks, Concept of Virtual circuit networks, structure of circuit and packet switch. Concepts of DSL and ADSL.	**11 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, students will be able to

CO1 - Understand the fundamental concepts of Data Communication and its components.
CO2 - Illustrate the digital and analog modes of transmission and its conversion.
CO3 - Discuss and select appropriate signal encoding technique for transmission.
CO4 - Analyze the data communication and carrier equipment.
CO5 - Compare transmission media based on transmission impairments and channel capacity.
CO6 - Examine multiplexing and switching techniques for signal transmissions.

TEXT/REFERENCE BOOKS

1. Data Communications and Networking, Behrouz Forouzan, McGraw-Hill.
2. Data and Computer Communications, William Stallings, Prentice Hall.
3. Computer Networks, Andrew Tanenbaum, Prentice Hall.